

Experiment No. 3

Title: Implement Control Structures and Form Validation; Create a Grading System Based on User-Entered Marks

Software / Tools Required:

1. Visual Studio Code
2. Google Chrome
3. HTML5
4. JavaScript (ES6)

Theory: JavaScript provides control structures that help in decision-making and repetition of tasks. Form validation ensures that user inputs are correct before submitting the data.

a) Control Structures

1. Used to control the flow of program execution.
2. Includes if, if-else, else-if, switch, for, while, and do-while.

b) Form Validation

1. Checks whether user input is valid.
2. Prevents incorrect or incomplete data from being submitted.

c) Conditional Statements

1. Used to compare values and execute specific blocks of code.
2. Helpful in implementing grading systems and validations. Case Study

Implemented:

A password validation webpage was created using JavaScript to verify whether the entered password is valid or not. A grading system was also designed where the user enters marks and the corresponding grade is displayed automatically.

Conclusion:

The practical was completed successfully. Control structures and form validation were implemented using JavaScript. A password validation system and a grading system were developed to validate user input and display the appropriate grade based on the entered marks.

Agentic AI Course: x Microsoft Word - A: x AI Coding Agents: L x AI Agent App Ideas: x Student Grading Sy: x + Ask Gemini

C:/Users/Mohammed/Downloads/JS%20Lab/exp3/index.html

Mathematics	Science	English
80	97	97

Calculate Grade

Result Card

Name: Mohammed
Roll No: 1042

Total Marks: 274 / 300
Percentage: 91.33%
Grade: A+

Excellent work! Keep it up.

Windows Taskbar: Chrome, Edge, Word, VS Code, etc. | System Tray: ENG, 9:03 AM

Agentic AI Course: x Microsoft Word - A: x AI Coding Agents: L x AI Agent App Ideas: x Student Grading Sy: x + Ask Gemini

C:/Users/Mohammed/Downloads/JS%20Lab/exp3/index.html

Student Grading System

Enter student details and marks

Student Name
Mohammed

Roll Number
1042

Subject Marks (out of 100)

Mathematics	Science	English
80	97	97

Calculate Grade